A PROOF AND LEAN FORMALIZATION OF THE CARLET ZERO-SUM SUBSPACE CONJECTURE

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Abstract. Let K be a finite field of cardinality 2^n, viewed as an n-dimensional vector space over $\mathbb{F}_2$. We prove the full odd- and even-degree ranges of the Carlet zero-sum subspace conjecture: if n is odd with $3 \le k$ and $k+3 \le n$, or n is even with $2 \le k$ and $k+2 \le n$, then there is a k-dimensional $\mathbb{F}_2$-linear subspace $E \subseteq K$ such that

$$\sum_{x \in E \setminus \{0\}} x^{-1} = 0.$$

Both parity ranges have been formalized in Lean 4 with universal quantification over the corresponding finite fields. The odd-degree proof uses root-product polynomials, Hasse bounds, low-dimensional constructions, and energy iteration; the even-degree proof uses two- and three-dimensional base cases, dimension recurrence, and complementary-dimension symmetry. The complete default build, independent verification, and final axiom audit pass: there are no compiler warnings, source and OLean hashes match, and the audited dependencies are only `propext`, `Classical.choice`, and `Quot.sound`, with no `sorryAx` or custom axiom.

1. Introduction and main theorem

Let K be a finite field of characteristic 2. Its nonzero elements form a finite group, but the inverse map $x \mapsto x^{-1}$ is not additive. This makes inverse sums on additive subspaces a natural finite-field problem: for which dimensions must a subspace contain a balanced collection of inverses? We study the zero-sum condition arising in Carlet's work [1], uniformly over all finite fields of a fixed cardinality.

For an $\mathbb{F}_2$-linear subspace $E \subseteq K$, write

$$S(E) = \sum_{x \in E \setminus \{0\}} x^{-1}.$$

We state the odd-degree range first, give the even-degree range separately, and then combine them into a parity-dependent conclusion. The Lean formalization preserves the original quantifiers in both cases.

Definition 1.1. A finite-dimensional $\mathbb{F}_2$-linear subspace $E \subseteq K$ is a *zero-sum subspace* if

$$S(E) := \sum_{x \in E \setminus \{0\}} x^{-1} = 0.$$

Theorem 1.2 (Odd-degree Carlet statement)**.** *Let $n, k \in \mathbb{N}$, with n odd, $3 \le k$, and $k+3 \le n$. For every finite field K of characteristic 2 and cardinality 2^n, there exists a k-dimensional $\mathbb{F}_2$-linear subspace $E \subseteq K$ such that $S(E) = 0$.*

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The assertion is uniform in K: the formal theorem quantifies over every field structure, finiteness structure, and $\mathbb{F}_2$-algebra structure satisfying the cardinality equality. The lower-range condition $k+3 \le n$ is retained exactly. The even-degree assertion is stated separately below with its own base dimension and admissibility condition.

2. Root products and a coefficient criterion

For a finite $\mathbb{F}_2$-subspace $E \subseteq K$, set

$$P_E(T) = \prod_{u \in E} (T+u).$$

This is a linearized polynomial. If $r = \dim_{\mathbb{F}_2} E$, then

$$P_E(T) = c_0 T + c_1 T^2 + c_2 T^4 + \cdots + T^{2^r}, \qquad c_0 \neq 0.$$

Lemma 2.1 (Reciprocal-sum criterion)**.** *For every nonzero finite-dimensional $\mathbb{F}_2$-subspace E,*

$$S(E) = \frac{c_1}{c_0}, \qquad \textit{and hence} \qquad S(E) = 0 \iff [T^2]P_E = 0.$$

Proof. Separate the zero root:

$$P_E(T) = T \prod_{u \in E \setminus \{0\}} (T+u).$$

The constant coefficient of the product on the right is $\prod_{u \neq 0} u = c_0$. Its linear coefficient is

$$\sum_{u \neq 0} \prod_{v \neq 0, u} v = \left(\prod_{v \neq 0} v \right) \left(\sum_{u \neq 0} u^{-1} \right) = c_0 S(E).$$

After multiplication by T, this becomes the T^2 coefficient of P_E. Since $c_0 \neq 0$, the equivalence follows. □

Let $I(E) = P_E(K)$ and let $D(E) = I(E)^{\perp}$ with respect to the trace pairing. Then $\dim_{\mathbb{F}_2} I(E) = n - \dim_{\mathbb{F}_2} E$ and $\dim_{\mathbb{F}_2} D(E) = \dim_{\mathbb{F}_2} E$. If $\dim_{\mathbb{F}_2} E = k$ and $a_{k-1} = [T^{2^{k-1}}]P_E$, the formalized dual-polynomial identity gives

$$S(D(E)) = a_{k-1}^2. \tag{2.1}$$

Thus a k-dimensional space whose root product has vanishing next-to-leading Frobenius coefficient yields a k-dimensional zero-sum space. In general this zero-sum space is $D(E)$, not necessarily E itself.

If $x \notin E$, grouping the roots into the two cosets of E and using $P_E(T+x) = P_E(T) + P_E(x)$ gives the extension identity

$$P_{E+\langle x \rangle}(T) = P_E(T)^2 + P_E(x) P_E(T). \tag{2.2}$$

This formula drives the low-dimensional constructions.

Lemma 2.2 (Complementary dimensions)**.** *If $\dim_{\mathbb{F}_2} K = n$ and $|K| > 2$, then K has a k-dimensional zero-sum subspace if and only if it has an $(n-k)$-dimensional zero-sum subspace.*

Proof. The linear map $P_E : K \to K$ has kernel E and image $I(E)$ of dimension $n-k$. The subspace polynomials satisfy

$$P_{I(E)}(P_E(T)) = T^q - T.$$

Write $c_0 = [T]P_E \neq 0$, $c_1 = [T^2]P_E$, $d_0 = [T]P_{I(E)} \neq 0$, and $d_1 = [T^2]P_{I(E)}$. Comparing T^2 coefficients gives $d_0 c_1 + d_1 c_0^2 = 0$, so $c_1 = 0$ if and only if $d_1 = 0$. Lemma 2.1 now shows that E is zero-sum exactly when $I(E)$ is. Since $\dim_{\mathbb{F}_2} I(E) = n-k$, this proves one direction; applying the same dimension relation once more gives the reverse implication. □

3. The three-dimensional case: an elliptic curve

Write $q = |K| = 2^n$ and define the binary Kloosterman sum

$$\mathrm{Kl}_K = \sum_{v \in K^\times} (-1)^{\mathrm{Tr}(v+v^{-1})}.$$

Consider the characteristic-two Weierstrass curve

$$C: \quad y^2 + xy = x^3 + 1.$$

Its discriminant is 1, so it is elliptic. Counting affine points by fibers over x and using the fact that $z^2 + z = c$ has $1 + (-1)^{\mathrm{Tr}(c)}$ roots gives

$$\#C(K) = q + 1 + \mathrm{Kl}_K\,.$$

The Hasse bound [2] therefore yields $|\,\mathrm{Kl}_K\,| \le 2\sqrt{q}$.

Partition the three-dimensional subspaces containing 1 by their next Frobenius coefficient. The exact finite counting identity formalized in the project is

$$(q-2)(q-4)\,p_3(0) = q + 1 - 3\,\mathrm{Kl}_K, \tag{3.1}$$

where $p_3(0)$ is the proportion of the corresponding finite family for which that coefficient vanishes. For odd $n \ge 7$, we have $q \ge 128$ and

$$q + 1 - 3\,\mathrm{Kl}_K \ge q + 1 - 6\sqrt{q} > 0.$$

Both factors on the left of (3.1) are positive, so $p_3(0) > 0$. A three-dimensional subspace with vanishing next coefficient exists; the coefficient-duality theorem and Lemma 2.1 turn it into a zero-sum subspace.

Proposition 3.1. *For every odd $n \ge 7$, every finite field K of cardinality 2^n contains a three-dimensional zero-sum subspace.*

4. The four-dimensional case: Fourier counting on a norm-one conic

We prove the four-dimensional case for every admissible odd degree. The count must impose two trace-orthogonality conditions at once, so that two values lie in the image of a chosen subspace polynomial.

4.1. **A two-dimensional seed and the extension identity.** Choose $t \in K \setminus \{0, 1\}$ and put $H = \langle 1, t\rangle_{\mathbb{F}_2}$. Since n is odd, $T^2 + T + 1$ has no root in K, so $c = t^2 + t + 1 \neq 0$. Direct multiplication gives

$$P_H(T) = T^4 + cT^2 + (t^2 + t)T.$$

Consider the norm-one conic

$$\mathcal{C}(K) = \{(a, u) \in K^2 : a^2 + au + u^2 = 1\}. \tag{4.1}$$

For odd n, this conic has $q+1$ points. Define the linear map $L(x) = c^{-2}P_H(x)$ and let $D = (\mathrm{im}\, L)^\perp$, where orthogonality is with respect to the nondegenerate trace pairing $(r, s) \mapsto \mathrm{Tr}(rs)$. Since $\ker L = H$, we have $\dim_{\mathbb{F}_2} \mathrm{im}\, L = n - 2$, $\dim_{\mathbb{F}_2} D = 2$, and $D^\perp = \mathrm{im}\, L$. Consequently, whenever $a, u \in D^\perp$, there are $x, y \in K$ such that

$$P_H(x) = c^2 a, \qquad P_H(y) = c^2 u.$$

Impose on a point of $\mathcal{C}(K)$ the two conditions $u \in D^\perp$ and $a + u \in D^\perp$, equivalently

$$\mathrm{Tr}(ru) = 0, \quad \mathrm{Tr}(s(a + u)) = 0 \qquad (r, s \in D).$$

Write $\chi(z) = (-1)^{\mathrm{Tr}(z)}$. For a nonzero frequency pair (A, B), the Fourier coefficient is

$$\mathcal{A}(A,B) = \sum_{x\in K} \chi\left(\frac{Bx^2+A}{x^2+x+1}\right) + \chi(B).$$

It is obtained by counting the fibers of the Artin–Schreier equation

$$\delta^2+\delta = \frac{(A+B)x^2+Ax}{x^2+x+1}.$$

A birational change of variables sends the generic solutions to the Weierstrass curve

$$y^2+xy+By = x^3+(A^2+B+1)x^2+(A+(A^2+B))x+B(A+(A^2+B))+A^2.$$

Its discriminant is $(A^2+AB+B^2)^2$. Odd extension degree ensures that T^2+T+1 has no root in K and that this discriminant is nonzero for every nonzero frequency pair. The Hasse bound controls the total curve-point count; the map from generic points to affine equation solutions is injective, and the exceptional fibers together account for at most five points. It follows that each nonzero-frequency coefficient satisfies $\mathcal{A}(A,B) \geq -2\sqrt{q}-4$. Using the four elements of D as additive-character filters for each condition yields a 16-term Fourier expansion. The zero-frequency term is $q+1$, while the other 15 terms satisfy the bound above. Thus the number N of points satisfying both conditions satisfies

$$16N \geq q+1-30\sqrt{q}-60. \tag{4.2}$$

If $q \geq 2048$, the right-hand side is greater than 48, so $N > 3$.

For every point on the conic, $a = 0$ forces $u = 1$, while $a = 1$ forces $u = 0$ or $u = 1$. There are at most three points with $a \in \{0, 1\}$. Choose a good point with $a \neq 0, 1$. Since $u, a+u \in D^\perp$, also $a \in D^\perp$, so the preimages x, y above exist.

Set $V = H + \langle x\rangle$. Then $\dim_{\mathbb{F}_2} V = 3$, and (2.2) gives

$$[T^4]P_V = c^2 + P_H(x) = c^2(1+a) \neq 0.$$

Furthermore,

$$\begin{aligned} P_V(y) &= P_H(y)^2 + P_H(x)P_H(y) \\ &= c^4(u^2+au) = c^4(1+a^2) = \big(c^2(1+a)\big)^2. \end{aligned}$$

Here we used $a^2+au+u^2 = 1$ and $(1+a)^2 = 1+a^2$ in characteristic two. The displayed nonzero value implies $y \notin V$. Extending once more by y, identity (2.2) shows that the T^8 coefficient of the resulting four-dimensional root product vanishes. The same-dimension coefficient-duality theorem then gives a four-dimensional zero-sum subspace.

Proposition 4.1. *For every odd $n \geq 11$, every finite field K of cardinality 2^n contains a four-dimensional zero-sum subspace.*

The remaining admissible degrees $n = 7, 9, 11, 13, 15$ are handled by exact finite-field constructions formalized in the project. Combining them with Proposition 4.1 gives the full base family:

Theorem 4.2. *For every odd $n \geq 7$ and every finite field K of cardinality 2^n, there is a four-dimensional zero-sum subspace.*

5. The five-dimensional case: a three-dimensional seed and a norm-one count

5.1. **A nonzero-coefficient seed from dimension four.** By Theorem 4.2 and coefficient duality, choose a four-dimensional subspace W with $[T^8]P_W = 0$. Let U be a three-dimensional hyperplane of W, and take $v \in W \setminus U$. Put $b = [T^4]P_U$. Comparing T^8 coefficients in (2.2) yields

$$0 = [T^8]P_W = b^2 + P_U(v).$$

Since the roots of P_U are exactly U, we have $P_U(v) \neq 0$, and hence $b \neq 0$. Put $c = b^2 \neq 0$.

Let $L = c^{-1}P_U$. Its image has codimension three. Set $D = (\operatorname{im} L)^\perp$; then $\dim_{\mathbb{F}_2} D = 3$ and $|D| = 8$. Impose on a point (a, u) of (4.1) the conditions $u, a+u \in D^\perp$. Filtering both conditions over the eight elements of D produces a 64-term Fourier expansion. The zero-frequency term is $q+1$, and each of the other 63 terms is at least $-2\sqrt{q}-4$. Therefore

$$64N \geq q + 1 - 63(2\sqrt{q}+4) = q + 1 - 126\sqrt{q} - 252. \tag{5.1}$$

For $q \geq 65536$, the right-hand side exceeds 192, hence $N > 3$. At most one point on the conic has $a = 0$, and at most two have $a = 1$. We may therefore choose a good point with $a \neq 0, 1$.

Because $u, a+u \in D^\perp = \operatorname{im} L$, their sum a also belongs to the image. Choose $x, y \in K$ with

$$P_U(x) = ca, \qquad P_U(y) = cu.$$

As $ca \neq 0$, we have $x \notin U$. Let $V = U + \langle x \rangle$, so $\dim_{\mathbb{F}_2} V = 4$. Write $A = [T^8]P_V$. Since P_U is monic of degree eight, (2.2) gives

$$A = b^2 + P_U(x) = c(1+a) \neq 0.$$

The conic equation implies $u^2 + au = 1 + a^2$. Thus

$$P_V(y) = P_U(y)^2 + P_U(x)P_U(y) = c^2(u^2 + au) = c^2(1+a^2) = A^2 \neq 0.$$

So $y \notin V$. With $Z = V + \langle y \rangle$, formula (2.2) yields

$$[T^{16}]P_Z = A^2 + P_V(y) = 0.$$

The five-dimensional coefficient-duality theorem converts this vanishing into a five-dimensional zero-sum subspace. This conversion is essential: the T^{16} coefficient is not itself the reciprocal sum; the proved same-dimension duality connects them.

Proposition 5.1. *For every odd $n \geq 17$, every finite field K of cardinality 2^n contains a five-dimensional zero-sum subspace.*

For the other admissible odd degrees $n = 9, 11, 13, 15$, exact finite-field constructions are formalized using explicit field models, linear-algebra witnesses, and a proof that the result transfers to every field of the same cardinality. Hence:

Theorem 5.2. *For every odd $n \geq 9$ and every finite field K of cardinality 2^n, there is a five-dimensional zero-sum subspace.*

6. Large dimensions: an actual probability recurrence and energy decay

Let Ω_d be the finite set of d-dimensional $\mathbb{F}_2$-subspaces containing 1. For $E \in \Omega_d$, let $b(E)$ denote its next Frobenius coefficient, and define the exact distribution

$$p_d(b) = \frac{\#\{E \in \Omega_d : b(E) = b\}}{\#\Omega_d}, \qquad f_d(b) = p_d(b) - q^{-1}.$$

Then $\sum_{b \in K} f_d(b) = 0$. The Lean development derives the recurrence for these distributions from the underlying finite-subspace counts; no abstract probabilistic model is substituted for the actual family.

For a nontrivial additive character $\psi : K \to \mathbb{R}$ and a real parameter s, define the transfer operator by

$$(T_s f)(b) = \frac{1}{q-s} \begin{cases} (q-s)f(0), & b = 1, \\ \displaystyle\sum_{v \neq 0} \left(1 + (s-1)\psi\left(\frac{b-1}{v}\right)\right) f(v), & b \neq 1. \end{cases}$$

Equivalently, the special branch is at shifted coordinate $b - 1 = 0$; this is the convention used in the implementation. Put $\mathcal{E}(f) = \sum_b f(b)^2$.

Lemma 6.1 (Two-step energy contraction)**.** *If $s \geq 2$ and $32s^2 \leq q$, then every zero-sum function f satisfies*

$$\mathcal{E}(T_{2s}(T_s f)) \leq \frac{4s^2}{q}\mathcal{E}(f).$$

If also $s^2 \leq q$ and $s < q$, then one step is nonexpansive: $\mathcal{E}(T_s f) \leq \mathcal{E}(f)$.

Proof. Decompose the transfer as a rank-one perturbation in the additive-character direction. Orthogonality of finite additive characters gives an exact squared-norm identity, including the exceptional coordinate term. For two successive ranks, the scalar on the distinguished direction is bounded by $2s/\sqrt{q}$; squaring gives $4s^2/q$. The condition $32s^2 \leq q$ makes the denominators $q-s$ and $q-2s$ positive and controls the remaining rank-one term. The same exact identity gives one-step nonexpansion when $s^2 \leq q$. The formal proof establishes the identity before applying these inequalities. □

The initial one-dimensional distribution is a point mass and has centered energy $1 - q^{-1}$. Iterating Lemma 6.1 along odd dimensions gives, for $k = 2r + 1$,

$$\mathcal{E}(f_k) \leq \frac{2^{2r^2+2r}}{q^r}(1 - q^{-1}).$$

For $k = 2r + 2$, apply one final nonexpansive step to obtain the same bound. If $p_k(0) = 0$, then $f_k(0) = -q^{-1}$, and this coordinate alone implies $\mathcal{E}(f_k) \geq q^{-2}$. Thus a strict upper bound below q^{-2} forces $p_k(0) > 0$.

For odd $k = 2r + 1$, the hypotheses $r \geq 5$ and $n \geq 4r + 3$ imply

$$2r^2 + 2r < n(r - 2),$$

so $2^{2r^2+2r}/q^r < q^{-2}$. For even $k = 2r + 2$, the same strict inequality follows from $r \geq 4$ and $n \geq 4r + 5$. Therefore the coefficient-zero probability is positive for odd $k \geq 11$ and even $k \geq 10$, whenever $n \geq 2k + 1$. A positive probability means an actual subspace has the desired vanishing coefficient; coefficient duality then gives a zero-sum subspace.

7. Even degree: base cases, recurrence, and complementary dimensions

The even-degree argument starts from different base cases. If n is even, K contains its unique subfield $\mathbb{F}_4$. Choose $\omega \in \mathbb{F}_4 \setminus \mathbb{F}_2$, so $\omega^2 + \omega + 1 = 0$. The sum of the inverses of its three nonzero elements is $1 + \omega + \omega^2 = 0$, giving the $k = 2$ base. For even $n \geq 6$, a three-dimensional base follows from the Kloosterman sum and the Hasse bound for an elliptic curve.

Theorem 7.1 (Even-degree statement)**.** *Let n be even and $|K| = 2^n$. If $2 \leq k$ and $k + 2 \leq n$, then K contains a k-dimensional zero-sum subspace.*

Proof. We first establish the three-dimensional base. Put $q = |K|$ and define $\mathrm{Kl}_K = \sum_{v \in K^\times} (-1)^{\mathrm{Tr}(v+v^{-1})}$. The point count for $y^2 + xy = x^3 + 1$ gives $|\,\mathrm{Kl}_K\,| \leq 2\sqrt{q}$. Partitioning the three-dimensional subspaces containing 1 by their next Frobenius coefficient yields the exact even-degree identity

$$(q-2)(q-4)p_3(0) = q - 13 + 3\,\mathrm{Kl}_K\,.$$

In even degree $\mathrm{Tr}(1) = 0$, so the character-weighted count differs from the odd-degree formula. For $n \geq 6$, we have $q \geq 64$, and the Hasse bound gives

$$q - 13 + 3\,\mathrm{Kl}_K \geq q - 13 - 6\sqrt{q} = (\sqrt{q} - 3)^2 - 22 \geq 3 > 0.$$

Since $(q-2)(q-4) > 0$, it follows that $p_3(0) > 0$. Thus a three-dimensional subspace with vanishing coefficient exists, and the coefficient criterion together with same-dimension duality gives a three-dimensional zero-sum subspace.

Now suppose $2k \le n$. Even dimensions start from the two-dimensional base, and each next case follows by adding a two-dimensional zero-sum subspace to one of dimension $k-2$. Odd dimensions start from the three-dimensional base just proved and use the same step. The dimension-addition lemma requires $2(k-2) < n$, which follows from $2k \le n$. This proves the result for every $2 \le k \le n/2$.

If $2k > n$, put $k' = n - k$. The assumption $k + 2 \le n$ gives $k' \ge 2$, and $2k' < n$. Apply the lower-half result to k' and then use complementary-dimension symmetry to obtain dimension k. When $n = 4$, the hypotheses allow only $k = 2$, which is the base case. □

8. Odd-degree dimension assembly

The low-dimensional cases also generate dimensions $6, 7, 8, 9$. Suppose E, F are zero-sum subspaces of dimensions k, ℓ and $k\ell < n$. A formalized linear-algebra lemma gives a nonzero scalar a with $aF \subseteq P_E(K)$. Define

$$G = P_E^{-1}(aF).$$

Since $\ker P_E = E$ and $\dim_{\mathbb{F}_2} P_E(K) = n - k$, we get $\dim_{\mathbb{F}_2} G = k + \ell$. The composition formula for subspace polynomials and the coefficient criterion show that G is zero-sum. Thus

$$\text{zero-sum dimensions } k, \ell \text{ and } k\ell < n \quad \Longrightarrow \quad k + \ell \text{ is a zero-sum dimension.} \tag{8.1}$$

Using dimensions 3 and 4, the pairs $(3,3), (3,4), (4,4), (3,6)$ give dimensions $6, 7, 8, 9$. Dimensions $k \ge 10$ are covered by the energy argument. Hence for the lower half $2k + 1 \le n$, the only base dimensions required are $3, 4, 5$.

For the upper half $2k + 1 > n$, put $k' = n - k$. Since n is odd and $k + 3 \le n$, one checks that $k' \ge 3$ and $2k' + 1 \le n$. Apply the lower-half result to k', then use Lemma 2.2 to obtain the desired k-dimensional space. This accounts for every admissible odd-degree parameter.

Proof of Theorem 1.2. If $2k + 1 \le n$, use Proposition 3.1, Theorem 4.2, or Theorem 5.2 for $k = 3, 4, 5$ respectively; use (8.1) for $6 \le k \le 9$; and use the energy argument for $k \ge 10$. If $2k + 1 > n$, reduce to $k' = n - k$ and apply complementary-dimension symmetry. The finite-field constructions and all transitions work for every K of cardinality 2^n, so the original universal quantifier is preserved. □

Theorem 8.1 (Combined odd- and even-degree statement)**.** *Let $n, k \in \mathbb{N}$, and let K be a finite field of characteristic 2 and cardinality 2^n. If*

$$(\mathrm{Odd}(n) \wedge 3 \le k \wedge k + 3 \le n) \quad \textit{or} \quad (\mathrm{Even}(n) \wedge 2 \le k \wedge k + 2 \le n),$$

then K contains a k-dimensional zero-sum subspace.

Proof. The odd case is Theorem 1.2; the even case is Theorem 7.1. Both hold uniformly for every finite field of the specified cardinality. □

9. Formal verification and reproducibility

The formalization uses Lean 4 and mathlib [3, 4]. The original odd-degree statement is defined as the proposition

$$\forall n, k, \quad \mathrm{Odd}(n) \to 3 \le k \to k + 3 \le n \to \forall K, \quad |K| = 2^n \to \mathrm{HasZeroSumSubspace}(K, k).$$

The original statement is defined in `Statement.lean`; the odd-degree theorem is `oddCarletConjecture_main` in `MainTheorem.lean`. The unconditional even theorem, `evenDegreeCarlet_unconditional`, is in `EvenDegree.lean`. It covers every even n, every finite field K with $|K| = 2^n$, and every k with $2 \le k$ and $k + 2 \le n$. Together these declarations imply Theorem 8.1. The main proof entry points are:

Proof component	Lean source
Reciprocal sum and root-product criterion	`Elementary.lean`, `Statement.lean`
Three-dimensional Hasse/Kloosterman argument	`KloostermanThree.lean`
All odd-degree four-dimensional cases	`K4AllOddDegreesProbe.lean` and imports
Five-dimensional norm-one construction	`K5NormOneConstructionProbe.lean` and imports
Distribution recurrence and energy contraction	`ProbabilityEnergy.lean`, `TwoStepEnergy.lean`
Dimension addition and complementary reduction	`AdditiveClosure.lean`, `LowDimensionReduction.lean`
Original quantified theorem	`MainTheorem.lean`
Even-degree bases and dimension recurrence	`EvenDegree.lean`, `KloostermanThree.lean`
Combined odd/even parameter ranges	`MainTheorem.lean`, `EvenDegree.lean`

The 7 November 2026 acceptance record uses Lean 4.33.0. The complete default build checked 270 modules; the independent verifier audited 923 declarations and checked 188 Hasse dependency modules. There were zero compiler warnings, and all source/OLean hashes match. Both main theorems passed independent statement checks. Their axiom sets contain only `propext`, `Classical.choice`, and `Quot.sound`; the audit found no `sorryAx` or custom unproved axiom. The Hasse bound is imported from the fixed AINTLIB snapshot [5] (`1c1c74664e40071c2c2165bc55ca2616a67ccd6b`); its source is `HasseBound.lean` and it is licensed under Apache 2.0. The pinned mathlib source baseline is `db584cd6d46c92f209a44c0f1c829460d327499d`. Reproduction commands are:

```
cd formalization
python -I build.py
python -I verify_build.py
python -I audit_final_theorem.py
```

The full-build summary and final axiom audit are recorded in `verified_build_summary.json` and `final_theorem_audit.json`.

The odd-degree proof follows the route checked by Lean: subspace polynomials, Hasse-controlled character sums, norm-one counts, and energy iteration. The even-degree proof uses two- and three-dimensional bases, dimension addition, and complementary-dimension symmetry. Finite computations are used only in the explicitly bounded small-degree branches, with the relevant constructions and transfer to arbitrary fields of the same cardinality checked in Lean.

AI-use statement

ChatGPT assisted with research organization, formalization support, and manuscript drafting. The human author is responsible for the mathematical content, verification record, and final text.

References

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[4] The mathlib Community, *The Lean Mathematical Library*, Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), pp. 367–381. DOI: 10.1145/3372885.3373824.

[5] C. Birkbeck, *AINTLIB*, Hasse–Weil bound module, project snapshot `1c1c74664e40071c2c2165bc55ca2616a67ccd6b` (Apache License 2.0).